

OAC Redstar common bean

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Smith, T. H., Michaels, T. E., Lindsay, A. M. and Pauls, K. P. 2009. **OAC Redstar common bean**. Can. J. Plant Sci. **89**: 309–311. OAC Redstar is a determinate bush dark red kidney bean (*Phaseolus vulgaris* L.) cultivar with early maturity and good yield potential. It is resistant to races 1 and 15 of bean common mosaic virus as well as race 17 of anthracnose. Its seed has acceptable cooking and canning quality.

Key words: *Phaseolus vulgaris* L., kidney bean, common bean, cultivar description

Smith, T. H., Michaels, T.E., Lindsay, A.M. et Pauls, K. P. 2009. **Le haricot OAC Redstar**. Can. J. Plant Sci. **89**: 309–311. OAC Redstar est un cultivar nain de haricot rouge foncé (*Phaseolus vulgaris* L.) à type de croissance définie, démontrant une maturité précoce et une bonne productivité. Il est résistant aux races 1 et 15 du virus de la mosaïque commune du haricot et à la race 17 de l'anthracnose. Le grain convient pour la cuisson et pour la mise en conserve.

Mots clés: *Phaseolus vulgaris* L., haricot rouge, haricot commun, description de cultivar

The dark red kidney bean (*Phaseolus vulgaris* L.) cultivar OAC Redstar (CFIA registration no. 6439) with a determinate growth habit, was developed by the Department of Plant Agriculture, University of Guelph, Guelph, Ontario. The cultivar was developed for production in traditional wide (70 cm) rows with direct combine harvesting. It is intended for production in areas of Ontario with more than 2650 crop heat units.

Pedigree and Breeding Methods

OAC Redstar was derived from the cross Montcalm/AC Litekid (Park and Tu, 1996b) //Cran09/Foxfire (Rogers Seed Company, Nampa Idaho) ///Chinook (Kelly et al. 1992)/AC Darkid (Park and Tu 1996a)//Generatif/OAC 90-C1, which was made in a growth room in 1994. The F₁ plants were grown in the field in 1995 at the Elora Research Station, Elora, Ontario, and all seeds were bulked. The F₂ to F₄ generations were advanced using bulk population advancement. F₂ and F₄ generations were grown in a winter nursery in New Zealand, while the F₃ generation was grown at the Elora Research Station in the summer of 1996. Single plant selections for early maturity, upright plant habit, high pod number and seed size, colour and shape were made from F₅ bulk population plots in the field at the Elora Research Station in 1997. The F_{5:6} generation was grown in 1998 at the Elora Research Station in progeny rows. The F_{5:7} generation was grown in preliminary yield trials in 1999 and the F_{5:8}–F_{5:10} generations were grown in advanced yield trials in 2000, 2001, 2002, all at the Elora Research

Station. Selection criteria for F_{5:6} to F_{5:9} populations included the same agronomic traits as for the F₅ population as well as high seed yield. Entries from advanced trials were entered into the Ontario Pulse Crop Committee's Coloured Bean Variety Registration Trials in 2003, 2004 and 2005. One hundred single plants from the F_{5:11} generation, grown in a growth room at the University of Guelph in the winter of 2005–2006, were rogued for uniformity and trueness to type, and the seed from the remaining plants was bulked and sent to Twin Falls, Idaho, USA, for breeder seed production in 2006.

OAC Redstar was tested for resistance to races 1 and 15 of bean common mosaic virus and races 17 (alpha) and 23 (delta) of anthracnose by artificial inoculation in growth rooms at the University of Guelph. Cooking quality was assessed for appearance by a panel at Agriculture and Agri-Food Canada Greenhouse and Processing Crops Research Centre, Harrow, Ontario. A Hunter Labscan (Hunter Associates Laboratories, Reston, VA) was used to determine dry and canned bean colour as *L* (white to black), *aL* (red to green), *bL* (yellow to blue) and *WI* (whiteness index). Canned bean texture was evaluated using the Ottawa texture measurement system for firmness (N mm⁻¹) and texture (N) (Voisey 1971). A visual examination for the degree of clumping of canned beans, the drain weight of washed-drained beans and the amount of water absorbed by a standard weight of beans were also used to judge cooking quality.

Table 1. Maturity, seed yield and seed weight of OAC Redstar and three checks in the Ontario Pulse Crop Committee White Bean Variety Registration Trials^z

Cultivar	Yield (kg ha ⁻¹)				Maturity (d)				Seed weight (g 100 seed ⁻¹)
	2003	2004	2005	Mean ^y	2003	2004	2005	Mean ^y	
OAC Redstar	2985	2603	2458	2674	92	98	92	94	59.0
Montcalm	2538	2249	2323	2358	98	102	96	99	57.8
AC Calmont	2931	2732	2711	2785	95	102	93	97	58.2
SE ^x	85.87	74.15	89.21	75.57	1.5	1.6	1.9	1.6	2.11
No. of trials	3	4	3	10	3	4	3	10	10

^zTrials were conducted at St. Thomas, Kippen, and Thorndale in 2003, in St. Thomas, Thorndale, Kippen and Monkton in 2004 and Kippen, St. Thomas and Thorndale in 2005.

^yBased on total number of trials.

^xBased on trial means of six cultivars.

Performance

OAC Redstar matured approximately 3 d earlier than AC Calmont and 5 d earlier than Montcalm. The yield was greater than Montcalm by 316 kg ha⁻¹ but less than AC Calmont by 111 kg ha⁻¹ over 10 trials in 3 yr (Table 1). The seed weight is similar to both check varieties. OAC Redstar is similar in both appearance and cooking quality to the check variety Montcalm, but with a higher plateau force and firmness than either check variety. OAC Redstar is resistant to races 1 and 15 of BCMV as well as race 17 of anthracnose, but is similar to Montcalm because it does not have the resistance to race 23 of anthracnose that is seen in AC Calmont (Table 2).

Other Characteristics

OAC Redstar has a green hypocotyl in the seedling stage and white flowers with pink stripes, which appear

approximately 42 d after planting. At maturity, plants were approximately 52 cm in height, which was 4 cm taller than AC Calmont. Pods at maturity are light tan in colour. The leaves are larger than AC Calmont or Montcalm. The green leaf colour is not as dark as Montcalm, but darker than AC Calmont. Seeds are dark red in colour, kidney shaped and slightly larger than both Montcalm and AC Calmont.

Maintenance and Distribution of Pedigreed Seed

Breeder seed of OAC Redstar is maintained by the Department of Plant Agriculture, University of Guelph, Guelph, Ontario, Canada N1G 2W1. Pedigreed seed is currently distributed through SeCan Association, 501–300 March Rd., Kanata, Ontario, Canada K2E 2E2.

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Table 2. Cooking quality and disease reaction of OAC Redstar and two checks in the Ontario Pulse Crop Committee White Bean Variety Registration Trials^z

Cultivar	Cooking quality						Disease reaction			
	Panel ^y	Texture ^x		Packing ^w	Drain wt. ^v	Hyd. Coef. ^u	BCMV ^t		Anthracnose ^s	
		Plat. (N)	Firm. (N mm ⁻¹)				1	15	17	23
OAC Redstar	3.2	224.23	18.86	3.0	66.22	1.92	–	–	–	+
Montcalm	3.2	172.90	17.53	3.0	67.60	1.86	–	–	–	+
AC Calmont	1.8	189.37	16.19	2.0	72.52	1.89	–	–	–	–
SE ^z	0.201	6.925	0.566	0.204	0.961	0.019				

^zTrials were grown at St. Thomas, Thorndale, and Kippen in 2005.

^yEight panelists evaluated for appearance of blanched beans canned in a salt-sugar brine using a scale of 1–5; 1, poor; 5, excellent.

^xTexture of washed-drained canned beans measured on Instron Texture measurement system using wire extrusion cells for plateau force (N) and firmness (N mm⁻¹).

^wPacking refers to degree of clumping of canned beans, scored visually from 1 to 5: 1, no clumping; 5, over half clumped.

^vDrain weight, weight of washed-drained canned beans expressed as a percentage of unwashed-undrained weight.

^uSoak weight/dry weight (determined for 500 g dry beans soaked for 40 min at 88°C).

^tResistance (–) or susceptibility (+) to bean common mosaic virus (BCMV) races 1 and 15.

^sResistance (–) or susceptibility (+) to Anthracnose races 17 and 23.

^zBased on trial means of seven cultivars.

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